

Visionary

WEIGHT 1 · Kanta Saito · BEHAVIOR / closed-loop / photoreal synth · 15+3

ONE-LINER

Treat Cosmos as BEHAVIOR's generative imagination + synth/eval layer beside $\pi_{0.5}$ motors and JEPA latent planners — hook is eval/augmentation/occlusion-fill, not drop-in CEM.

SPINE: Generative video as physics simulator — Predict2.5 (flow T2W/I2W/V2W) + Reason1 + Transfer2.5 | PARALLEL to latent WMs (Dreamer/JEPA/LeWM)
| synth data · policy eval · closed-loop robotics/AV

BEHAVIOR role — observation factory + eval harness

NOT the motor policy

Observation factory

Transfer2.5 Real2Real visual randomization (colors, lighting, distractors, backgrounds) for domain shift.

Eval harness

Roll policies into Predict/Transfer worlds; score visual success / VideoAlign / detectors before real trials.

Not motor policy

Winning stacks lean $\pi_{0.5}$ / FM action experts + strong VLM towers — Cosmos is eyes/synth, not torques.

24/30 maps to shift

In-paper evidence: Transfer augs beat classic image augs on visual OOD — moral for BEHAVIOR domain shift.

Label carefully — DreamGen IF $\neq$ BEHAVIOR SR

Tab.18 hygiene for Visionary bets

WHAT Tab.18 IS

Video instruction-following (Object GPT 91.8 / Behavior / Env) judged by VLMs on generated GR1 videos.

CLEAR DreamGen path: text→robot video→IDM/latent actions for VLA data scaling.

WHAT IT IS NOT

NOT BEHAVIOR-1K success rates. NOT hardware VLA task SR. Do not invent BEHAVIOR scores for Cosmos.

π_0 / OpenVLA not cited — rhyme as TENTATIVE transfer of synth-obs idea only.

Visionary discipline

Use DreamGen as a data-scaling recipe sketch; measure BEHAVIOR SR in follow-up work — never paste Tab.18 into a BEHAVIOR leaderboard slide.

Hybrid stack (TENTATIVE research — not in-paper)

$\pi_{0.5}$ /FM motor + LeWM/2-AC latent search + Cosmos synth/eval/slow imagination

Motor

$\pi_{0.5}$ / FM action experts

execute motor chunks

Latent search

LeWM / V-JEPA 2-AC

fast subgoal CEM in embedding

Generative layer

Cosmos Predict/Transfer

(a) synth demos (b) photoreal eval (c) slow rare-event imagination

TENTATIVE: research proposal for 2026 BEHAVIOR — label as Visionary, not Cosmos paper claim.

When generative WFM vs when latent

2026 BEHAVIOR decision table

Need	Prefer	Defer
Novel visual gen. for BC/VLA	Transfer2.5 Real2Real	Train Cosmos from scratch
Fast stage-level planning	LeWM / V-JEPA CEM	DiT video inside MPC
Task/scene diversity classic sim	RoboGen-style propose	Pixel WFM as task author
Photoreal eval / human prefs	Predict2.5 infer (2B first)	14B until needed
AV / indoor what-if maps	Transfer auto/multiview	Course smoke
Course proof	Sibling latent + RC1/2/4	PAI absolute / Tab.13 lab

Motion-planning / long-horizon / closed-loop hooks

Honest — no invented BEHAVIOR SR

RNDS long-horizon

Transfer chunked video stability (Fig.10) as backdrop for multi-step household scripts — still need a planner/policy.

AC chunks

Action-cond Predict2.5 for short-horizon what-if (Bridge metrics); complementary to 2-AC latent CEM — richer but slower.

Multiview / occlusion

Camera-controllable head→gripper synth fills occlusions for rearrange without moving physical cameras.

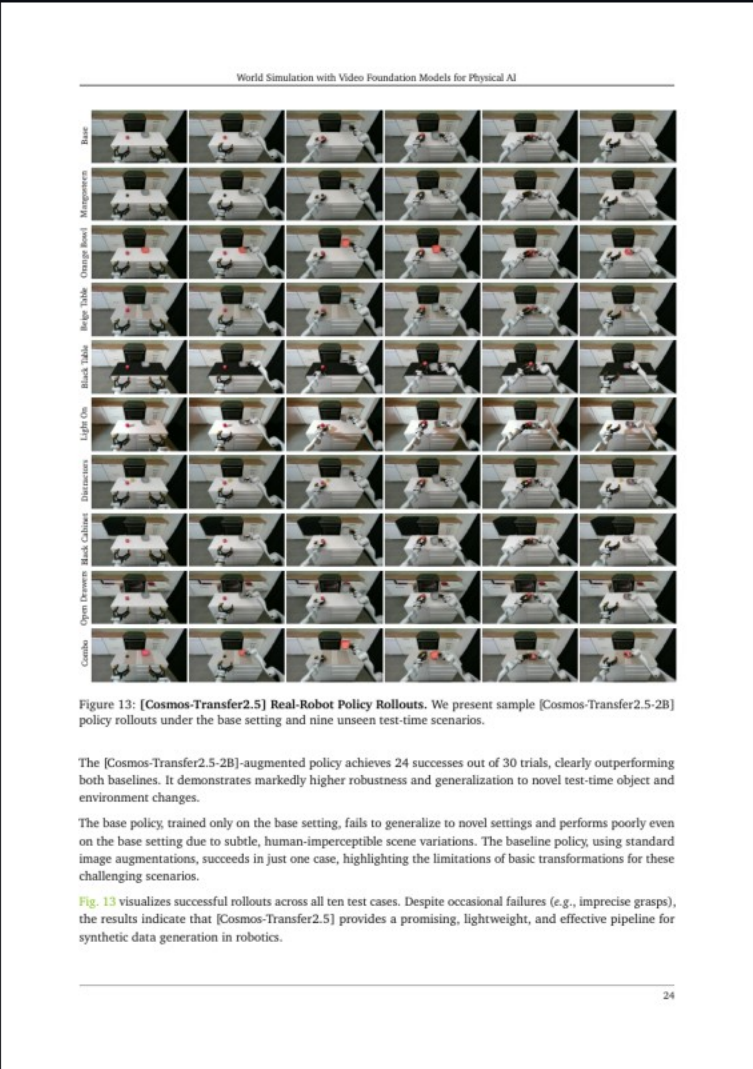
Generative eval harness Policy ↔ Transfer/Predict worlds at fixed Hz with distilled 4-step (robot Hz not reported — TENTATIVE).

Gap vs BEHAVIOR-1K

No household long-horizon language-goal SR in-paper; Tab.13 is short bimanual pick-place under visual OOD.

Fig. 11-13 — visual domain shift moral

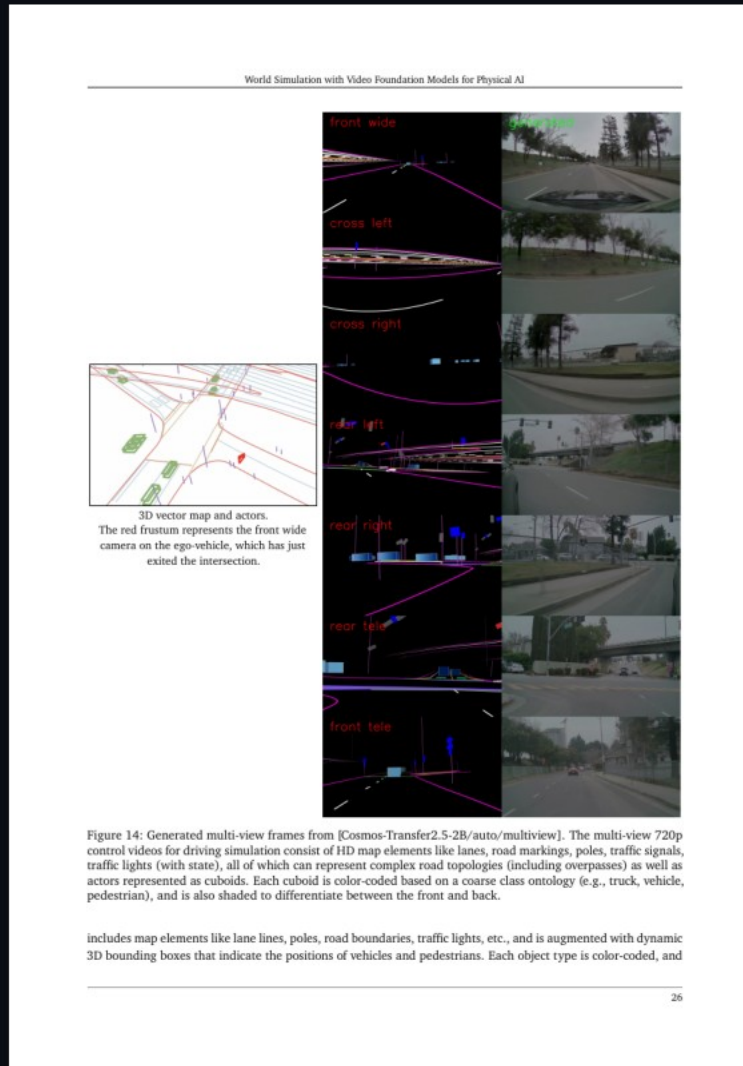
24/30 maps to BEHAVIOR's notorious appearance shift — not a BEHAVIOR score



Transfer Real2Real aug → Diffusion Policy. Offline synth, CLEAR robotics number.

Fig. 14 — multiview / world-scenario maps

AV recipe adaptable to indoor semantic floorplans (new app sketch)



Map/scenario → camera video. Indoor adaptation = Visionary, not paper claim.

Fig. 20 — AC Bridge policy-eval WM
Action-conditioned imagination for what-if — video metrics, not control SR table

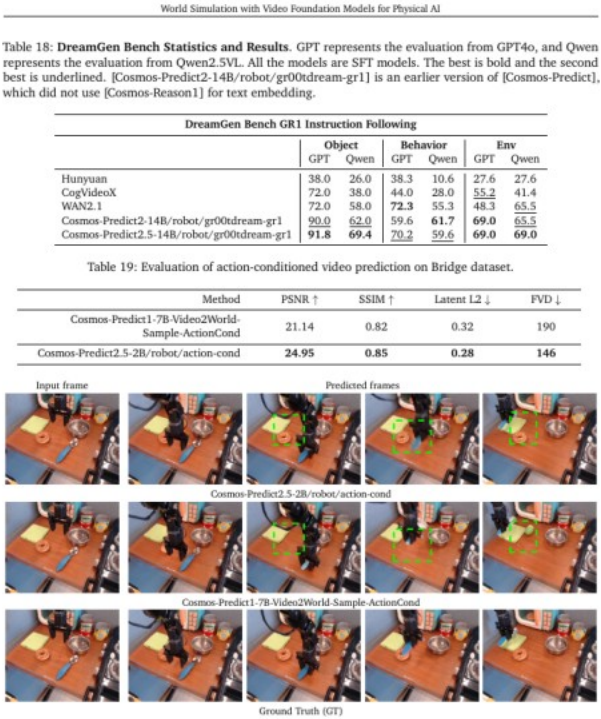


Figure 20: **Action-conditioned video prediction samples on the Bridge dataset.** Comparison of predicted rollouts from [Cosmos-Predict2.5-2B/robot/action-cond] and [Cosmos-Predict1.7B-Video2World-Sample-ActionCond] against the ground-truth frames. [Cosmos-Predict2.5-2B/robot/action-cond] demonstrates better object permanence. The green dotted box highlights the parts with the object permanence issues.

relative to the ground-truth frames.

We further investigate various methods for incorporating action conditioning. In addition to applying it through time embeddings, we also explore two alternatives: (1) cross-attention within the DiT blocks and (2) channel concatenation. The results are presented in Tab. 20.

Synthetic data recipe for FM / VLA stacks

Careful rhyme — π_0 not cited

1

Real teleop / OXE demos

seed distribution

2

Optional RoboGen tasks

classic sim diversity

3

Transfer2.5 appearance

Real2Real expand

4

Train DP / $\pi_{0.5}$ / FM

TENTATIVE consumer rhyme

5

Predict AC eval

imagination / regression tests

6

Real deploy

keep latent planner optional

Follow-up research agenda (Kanta)

No Discord / Drive — research only

BEHAVIOR visual OOD suite

Transfer2.5 prompts (materials, lighting, clutter) → measure VLA SR lift.

Closed-loop Hz study

Policy ↔ Transfer/robot multiview with distilled 4-step; report achievable Hz.

Hybrid tournament

Cosmos judge · LeWM/2-AC propose — generative-eval + latent-plan.

Physics-stress prompts

Paper's physics domain pack as BEHAVIOR safety scenarios.

Shared-episode bake-off

AC Cosmos rollouts vs V-JEPA 2-AC CEM on Bridge/DROID (video vs control).

New apps

Isaac/BEHAVIOR photoreal overlay · multi-cam wrists · kitchen visual unit tests
· indoor floorplan maps.

Visionary bets (weight 1)

Decision-oriented · Empiricist empty → no Tab.13 replication demand

Dual-WM stack

Latent planner PLUS generative synth/eval — don't pick one religion.

Cosmos as appliance

Data/eval appliance in BEHAVIOR; LeWM/JEPA as trainable curriculum.

Mix card

Real teleop : Transfer-aug : RoboGen-sim — leave-one-out later (other packs).

Non-goals

Students won't reproduce 4096-H100; success = correct tool choice + honest probes.

Fig. 9-10 — long-horizon Transfer stability

Closed-loop narrative ingredient (still need a policy)



Figure 9: Sample comparison results of [Cosmos-Transfer2.5-2B]. Compared to [Cosmos-Transfer1-7B], [Cosmos-Transfer2.5-2B] has better prompt alignment, better adherence to control input, and less hallucination and error accumulation (especially for long videos).

degrades across chunks. RNDS is defined as a curve over chunk indices:

$$RNDS[i] = \left(\frac{DOVER[i]}{DOVER_{GT}[i]} \right) / \left(\frac{DOVER[1]}{DOVER_{GT}[1]} \right), \quad (5)$$

where $i = 1, \dots, T$ denotes the chunk index, $DOVER[i]$ is the Dover score (Wu et al., 2023) of the i -th generated chunk, and $DOVER_{GT}[i]$ is the corresponding Dover score for the ground-truth video. This normalization ensures that the RNDS curve always starts at (1, 1), making it easy to compare degradation trends across models. The averaged RNDS is then obtained by averaging curves over all evaluation videos.

As shown in Fig. 10, the RNDS curves reveal that [Cosmos-Transfer2.5-2B] exhibits far less reduction in RNDS over time compared to [Cosmos-Transfer1-7B]. This indicates that our smaller model accumulates fewer errors, demonstrates less hallucination, and maintains higher fidelity over long video sequences.

6.2. Cosmos-Transfer2.5 for Robot Policy Learning

We aim to investigate the following question: Can [Cosmos-Transfer2.5-2B] be used as a visual synthetic data generator to augment robot policy training and enable generalization to unseen visual scenarios?

Our setup follows a standard real-world imitation learning pipeline. Using a bimanual robot equipped with an egocentric camera, we first collect human teleoperation demonstrations for table-top manipulation tasks. From these demonstrations, we train a vision-based policy that maps image observations and proprioception to action chunks using state-of-the-art behavioral cloning techniques. The trained policy is then deployed back on the same platform for evaluation.

Unlike conventional imitation learning benchmarks, however, we introduce adversarial visual perturbations during evaluation—for example, modifying object appearances, changing scene backgrounds, or placing distractor objects on the table. This setting reflects a realistic deployment scenario, where a policy must operate

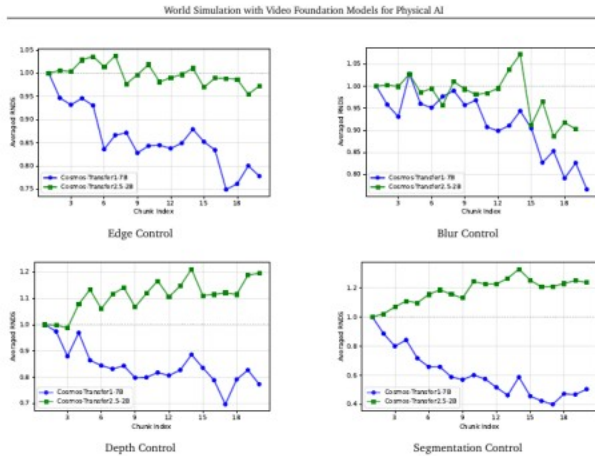


Figure 10: Error accumulation for long video generations. These plots show the Normalized Relative Dover Score vs Chunk Index for auto-regressive multi-trunk long video generation where each trunk is 93 frames. As shown, for all four control modalities (edge/bur/depth/seg), compared to [Cosmos-Transfer1-7B] (blue curves), [Cosmos-Transfer2.5-2B] (green curves) has much less reduction in RNDS along the chunk index dimension, which shows less hallucination and error accumulation for long videos.

in environments that differ drastically from the conditions in which the demonstrations were collected. Such domain shifts often involve structured visual changes that cannot be easily synthesized using standard image augmentation methods.

Here, [Cosmos-Transfer2.5-2B] offers a unique advantage: it not only enables the generation of these structured variations for visual data augmentation, but also provides controllability through text prompts that specify the desired visual conditions. This enables the systematic simulation of challenging out-of-domain scenarios and the testing of policy robustness in a controlled yet flexible manner.

6.2.1. System and Task Settings

We conduct our experiments on a semi-humanoid robotic platform equipped with two 7-DoF Kinova Gen3 arms, each fitted with a Robotiq 2F-140 gripper. An Intel RealSense D455 camera is mounted on the robot's head to capture egocentric image observations. The robot's base is fixed in front of a table and remains stationary throughout all experiments to ensure consistency.

For teleoperation, we use Meta Quest 2 controllers to track the 6D target poses of the left and right end effectors. These 6D poses are converted into target joint positions and velocities via a GPU-accelerated model predictive control (MPC) framework from cuRobo (Sundaralingam et al., 2023). The resulting commands are then executed by the robot's low-level joint impedance controller, enabling smooth and responsive teleoperation.

Numbers tattoo — pin these (KanBot must-cites)

Shared spine across all five seats · paper numbers only

24/30

Real2Real robot vs 5/30

0.768

PAI T2W Overall post

0.810

PAI I2W Overall post

24.95

Bridge AC PSNR

IF $\neq$ SR

DreamGen Tab.18 $\neq$ VLA

0.394

AV LET-AP Transfer2.5

3.5×

Transfer smaller vs T1

200M

curated video clips

2B / 14B

Predict scales

93f @16fps

~5.8 s generation

PARALLEL

not JEPA child

GENERATIVE

pixel WFM pole

Visionary — close

Takeaways + exam bite

1. Cosmos = BEHAVIOR observation factory + eval harness — not motor policy.
2. 24/30 maps to visual domain shift; DreamGen IF $\neq$ BEHAVIOR SR — label carefully.
3. Hybrid TENTATIVE: $\pi_{0.5}$ /FM + LeWM/2-AC + Cosmos synth/eval/slow imagination.
4. Hooks honest: RNDS, AC chunks, multiview — no invented BEHAVIOR scores; no Discord/Drive.

EXAM BITE

Explain Cosmos's BEHAVIOR role in one sentence and name one TENTATIVE hybrid component that is NOT an in-paper claim.